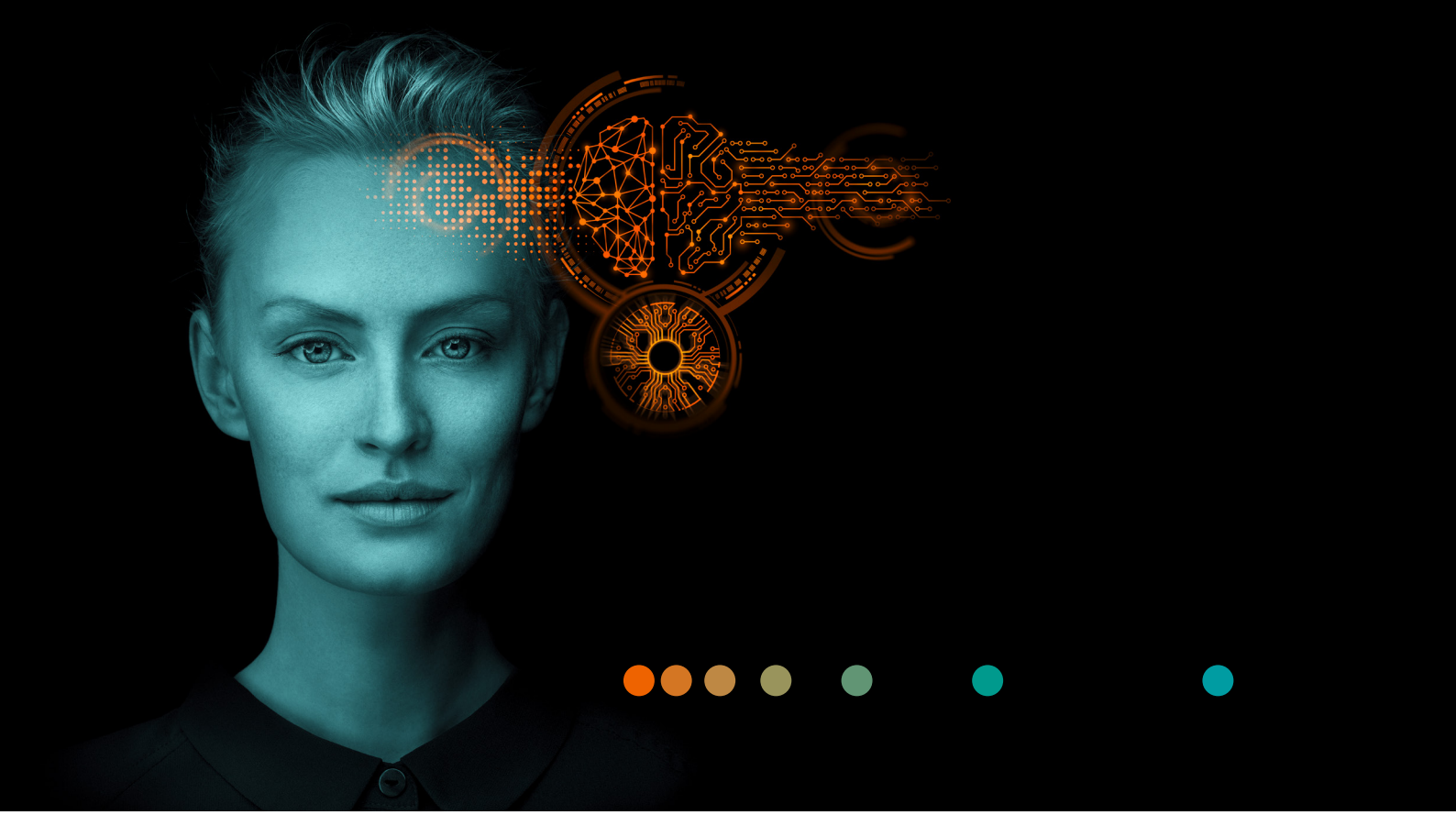


White paper

# Artificial intelligence – Advancing medicine with high-impact AI solutions

SIEMENS Healthineers

# Introduction

Artificial intelligence (AI) has been the talk of the town for some time now. Generative AI in particular has the potential to revolutionize healthcare and can change the approach to diagnosis, treatment and patient management. AI-supported solutions can help medical professionals make more accurate diagnoses, personalize treatment plans and optimize operational efficiency. Our company – Siemens Healthineers – has been working on the application of artificial intelligence in medicine for more than 20 years and has more than 1,100 patent families in the field of machine learning. We now have over 100 AI-enabled products on the global market and by 2025 there should be a total of 110 AI-supported products worldwide. According to the US Food and Drug Administration (FDA), we are one of the leading companies in terms of the number of FDA-approved AI-supported products[1].

This white paper is intended to showcase our approach and working method with artificial intelligence, as well as to illustrate our guidelines and security measures for data protection.

[1] *https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices*

## Contents

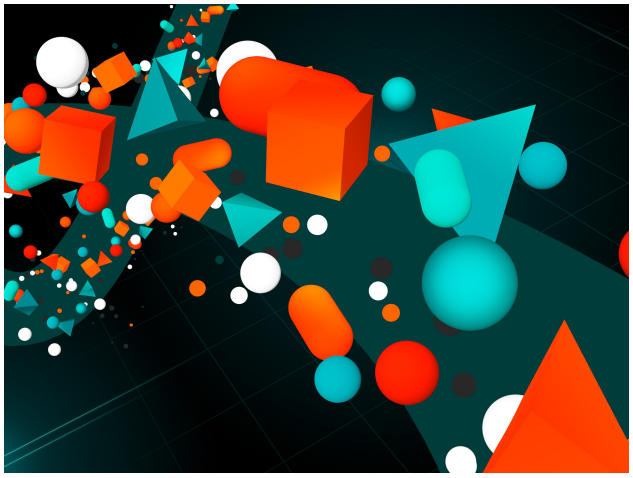

| | |
|---|---|
| **Introduction** | **2** |
| **Researching and developing artificial intelligence** | **3** |
| **Data lifecycle** | **5** |
| **Algorithms** | **6** |
| **Supercomputing** | **7** |
| **AI principles** | **8** |
| **Data privacy and protection** | **9** |
| **Collaboration** | **10** |

# Researching and developing artificial intelligence

## Advancing medicine with high-impact AI solutions

Siemens Healthineers is transforming big data into precision medicine. We're researching and developing scalable AI solutions that help automate the clinical routines of complex diagnostics and therapies. We call it AI factory and it's built on a reliable infrastructure that includes hardware, software, and experts.

## Big data is fueling innovation in healthcare



Real-world medical data is the cornerstone of AI innovation in healthcare. We harness the power of data responsibly and ensure its integrity and privacy so we can develop groundbreaking AI solutions that transform healthcare.

### Sourcing from all five continents

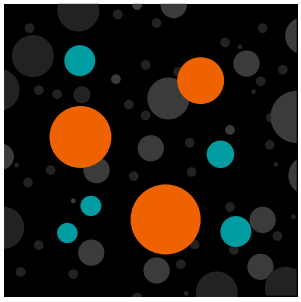
Access to comprehensive health data is essential for advancing artificial intelligence (AI) in healthcare. Siemens Healthineers meticulously gathers datasets from all five continents that encompass a diverse range of sources, including public clinical registries, medical associations, and trusted research partners. These datasets contain a wealth of information, including clinical images, lab data, genomic data, and clinical data like patient histories.

### Protecting patient privacy

Siemens Healthineers adheres to stringent data privacy principles to ensure that patient information remains secure and protected. Before data is used in research projects, it undergoes a rigorous anonymization process to remove all personally identifiable information. This safeguards patient privacy while enabling valuable research advances.

### Enhancing data quality

AI applications are fundamentally dependent on the quality of the data they're fed. Our experts thoroughly examine each data point, enriching it with additional information like anatomical landmarks, diagnostic indicators, and tumor characterizations. These annotations are crucial for developing accurate and reliable AI models that can transform healthcare.



## Two billion data points

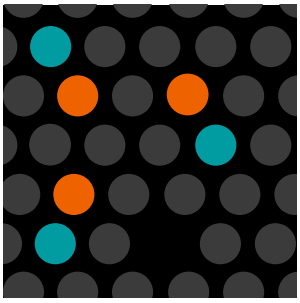
At the heart of our AI innovations lies a vast reservoir of data, a veritable data lake teeming with over two billion data points. This ever-expanding repository empowers us to continuously refine and enhance our AI algorithms.

## Insights rooted in robust data

With each new data point, our understanding of diseases and patient populations deepens, leading to increasingly intricate and comprehensive databases. This wealth of information is the bedrock of our AI development: It enables us to create groundbreaking solutions that transform healthcare.

## Comprehensive databases

AI thrives on vast amounts of high-quality data. Our AI models are enriched by a comprehensive array of data sources encompassing patient health status, medical results, clinical studies, and established clinical treatment standards defined by medical associations.



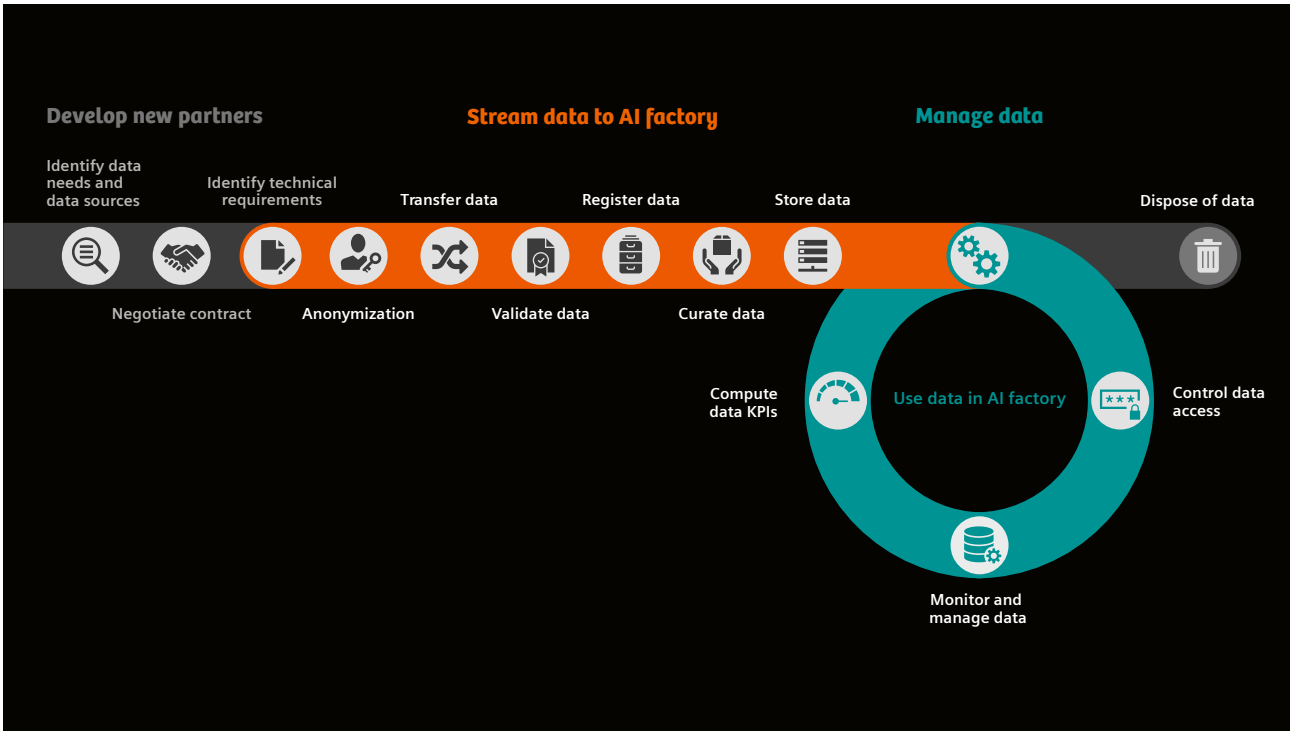
## Maximum data integrity and security

Prior to utilization in research endeavors, data must undergo a rigorous, standardized procedure to safeguard data integrity and security to the utmost extent. Each data point traces its origin to a patient, and our commitment to safeguarding their privacy rights remains unwavering. This is why all information capable of directly identifying an individual is thoroughly anonymized. Personal information is eliminated before the datasets even reach Siemens Healthineers.

# Data lifecycle

## The foundation of AI-driven healthcare

By leveraging global data sources, upholding data privacy, and prioritizing data quality, we're proving our commitment to fostering groundbreaking AI applications that revolutionize healthcare diagnostics, treatment, and patient outcomes.



# Algorithms

## Building algorithms for scalable AI-powered solutions

Creating and implementing algorithms in the field of healthcare is a meticulous process that involves seven key steps.

### Step one: Identifying medical needs

Algorithms can automate routine tasks, and that's a great asset when it comes to making healthcare more efficient and effective. We focus on developing scalable solutions that help physicians improve the quality of diagnostics and therapies, especially in the areas of cardiology, neurology, and oncology. To identify valuable innovations that can be integrated in clinical routines, we collaborate with leading medical institutions around the world.

### Step two: Accessing health data

AI relies on accessing and correlating a large amount of high-quality data. The data models incorporate factors like state of health, medical results, and clinical studies. Clinical standards for treatments defined by medical associations are also used to develop computational models and their architecture. Our work primarily focuses on key clinical indications like cardiovascular diseases, cancer, and neurological diseases.

### Step three: Developing a computational model

Algorithms are computational models created by a team of data engineers, medical scientists, and programming experts. Each application is uniquely designed to answer a specific question and automate its clinical routine process. Basically, it compares the (health) data with the probability of (clinical) outcomes. We hold over 1,100 patent families related to machine learning, with more than 550 of these pertaining to deep learning.

### Step four: Training the algorithm

We develop scalable AI-powered solutions, and that's why we maintain a robust infrastructure that includes supercomputing capabilities and a data lake. The quality and volume of the data used for training directly impact the accuracy and effectiveness of the algorithm's predictions.

### Step five: Validating the algorithm

The algorithm and its performance undergo a government-regulated testing and approval process. We also have to demonstrate the safety and benefits of the algorithm in patient treatment.

### Step six: Deploying the solution

The algorithm is embedded in a software solution or device. If the algorithm and its application have been validated, it can be implemented and used in clinical routines.

### Step seven: Monitoring and ongoing development

The final step involves monitoring the performance of the AI in clinical routines and updating the computational models as medical advances are made. This ensures that the algorithms continue to perform optimally and remain relevant in the ever-evolving field of medicine.

# Supercomputing

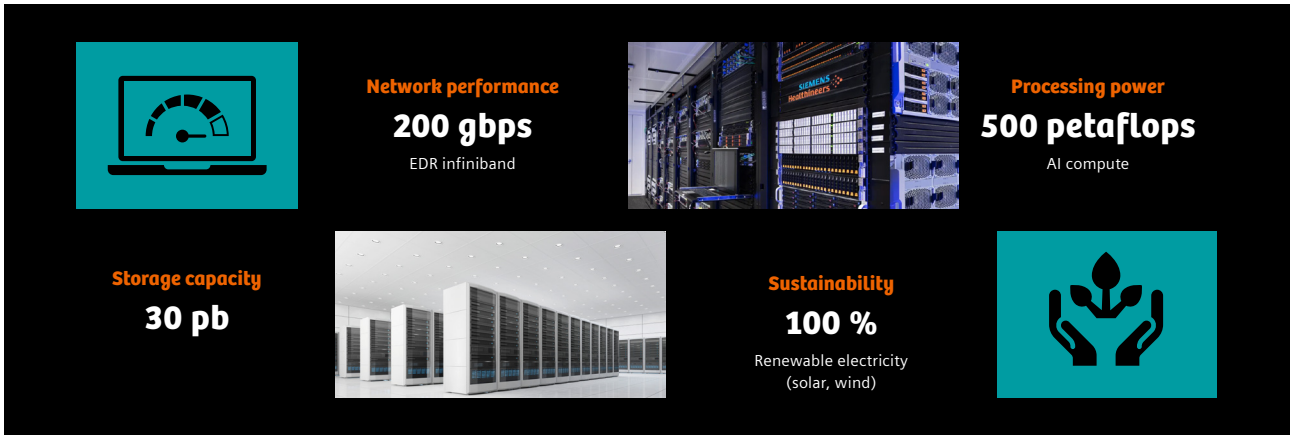
## A reliable and sustainable infrastructure is the backbone of our AI factory

At the heart of our AI development is a team of over 350 highly skilled experts who meticulously develop, train, and validate our AI algorithms. To support their work, we've established a global computing network that provides the infrastructure needed for AI experimentation.

**Our experts conduct over 1,600 AI experiments every day**

Our flagship supercomputer Sherlock is one of the most powerful AI development platforms in the healthcare industry. Its exceptional computing capacity allows our experts to conduct over 1,600 AI experiments every day and drive continuous progress in AI-powered healthcare solutions. Sherlock is operational in the United States and is supported by our decentralized infrastructure in China, Germany, and India.

**Global network** **Regional hubs** **Scientific collaborations**



# AI principles

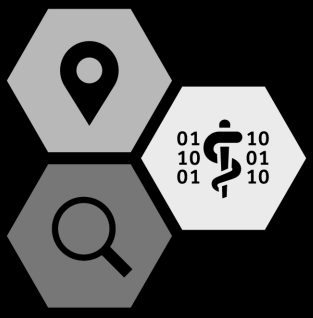
## Our purpose makes us a strong team of experts

We've devised nine principles that guide the way we develop and implement artificial intelligence in the field of healthcare.

We develop AI solutions to **support patients' desires** for **more personalized medicine.**

We believe **healthcare professionals** together with **AI solutions** are **a strong team.**

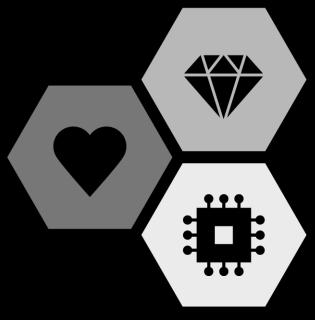
We believe the **level of autonomy of AI solutions** needs to be **balanced** with **ethical expectations and human values.**




We work passionately to **make AI solutions accessible** to patients **everywhere.**

We believe **data handling in healthcare** needs to **focus** on the **individual.**

We believe **AI development** needs to be **transparent.**



We **measure ourselves** against **highest scientific standards.**

We strive to develop **AI solutions** for **the healthy and the sick.**

We **speak honestly** about the **capabilities** of our **AI solutions.**

# Data privacy and protection

## Our AI factory is run by a strict corporate policy

Data privacy and data protection are at the core of our business. Over and above government regulations, we've also developed our own corporate policy.

**We use data for the benefit of the individual.**

The purpose of our company is to advance human health. People should benefit from data-driven medical innovations that prevent disease and provide best-in-class procedures and treatment. We invest in data-driven health solutions because we support the patient's desire for personalized, high-precision medicine that allows them to live a longer and healthier life.

**We use data to drive healthcare innovation.**

Data will become the key enabler for innovations in digital healthcare. Data-driven innovations are essential for medical research and progress. Our tailored and responsible use of data enables us to fill our innovation pipeline, push data-driven medicine, and develop innovative procedures for patients.

**We're trustworthy and ethical in our handling of data.**

We only use data in a purpose-bound manner to develop medical innovations and to enable our data-driven products to perform according to their specified performance capabilities. We treat data responsibly, reliably, and securely.

**We apply strict and proven data privacy standards worldwide.**

We believe that trust and accountability lay the foundation of responsible data privacy management, and therefore we apply high data privacy standards worldwide. The fundamental legal principles of the GDPR – including legitimacy and lawfulness of data processing, purpose limitation, the need-to-know principle, data avoidance, and data economy – are mandatory for Siemens Healthineers worldwide based on our own internal directives. In addition, we apply proven technical standards and organizational measures to ensure data security, authenticity, and confidentiality. Our ISO-certified cybersecurity management system follows a holistic approach and integrates information security management (ISO 27001) and privacy information management (ISO 27701).

**We support the advances that enable individuals to have transparent health data and sovereignty over that data.**

Every person should have sovereignty over their own health data. This includes being transparent about what data is used, how it's used, and for what purposes. They also need to have the right to grant or revoke consent to the use of their data. This right should include the freedom to donate their personal data for the purpose of conducting research, advancing the field of medicine, and improving healthcare solutions.

The processing of health data in private-sector research and development work also makes a significant contribution to advancing medical and technical progress. To safeguard this valuable contribution, we believe that private-sector research should also be subject to the privilege of research, and that developing medical devices and artificial intelligence that facilitate improvements in the early detection and treatment of illnesses, among other things, also serves the public interest and our public health.

We promote trust throughout society and among all patients in our application of digital technologies, and we support them in exercising their rights accordingly.

**We leverage data as a strategic asset.**

Driving digitalization and promoting value creation from data are essential to advancing medical progress and providing efficient, high-quality healthcare. Leveraging the potential of data is strategically important to us. In addition to developing data- and software-driven solutions that support decision-making, we work to continuously develop our portfolio by automating devices and workflows and expanding our use of predictive maintenance. The interoperability and connectivity of our products and solutions are accelerating this development into a platform-oriented business.

**We use state-of-the-art technologies to protect data.**

We offer a state-of-the-art portfolio of secure products, cybersecurity services, and consulting that helps ensure optimal protection. We continuously improve our systems and processes and train our teams on cybersecurity and data protection to maintain a consistently high level of threat awareness. Our engineering practices include a secure development lifecycle (SDL) to ensure that high cybersecurity standards are implemented for every product and solution. Examples of our core development principles include the implementation of privacy by design and privacy by default.

**We support open standards for data interoperability.**

The key to data-driven healthcare innovation is the ability to interconnect various health datasets. It's only through data integration and data interoperability that the value of data can be fully utilized. We strongly support the standardization of healthcare data and data sharing. When designing our solutions, we aim to systematically include standardized interfaces like DICOM, FHIR, and increasingly uniform APIs.

**We invest in trustworthy partnerships to access data.**

Efforts to improve medical knowledge and data-driven healthcare solutions depend largely on having the right to access health data from diverse and genuine sources. We believe that providing fair access to relevant data to all healthcare stakeholders and using this data responsibly to our mutual benefit will contribute to advancing the field of medicine. We therefore build our data-related partnerships on a foundation of fairness and transparency.

# Collaboration

## Developing breakthroughs in trusted partnerships

### Strategic focus on tackling big challenges

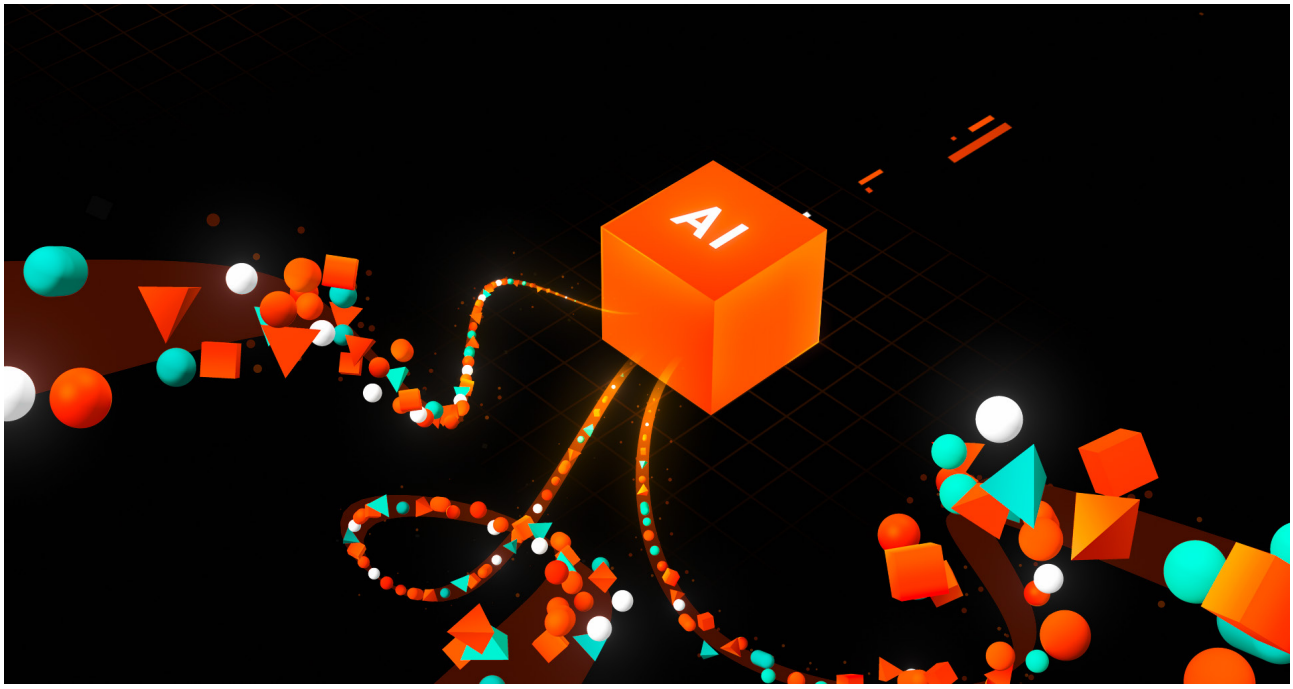
We strategically focus our efforts on key clinical indications, including cardiovascular diseases, cancer, and neurological disorders, which enable us to develop AI solutions that address some of the most pressing healthcare challenges.

### Dynamic ecosystem of knowledge and expertise

Our partnerships are the cornerstone of our commitment to delivering groundbreaking healthcare innovations. By collaborating with over 4,500 esteemed partners, including renowned medical centers, hospitals, and university hospitals, we foster a dynamic ecosystem of knowledge and expertise.

### Unwavering dedication to collaborative innovation

This global network allows us to engage in approximately 150 research and development projects per year, which is a testament to our unwavering dedication to collaborative innovation. Our trusted partnerships aren't just alliances, they're the lifeblood of our AI development process because they provide us with critical data, research insights, and diverse perspectives necessary to advance AI in healthcare. Along with our partners, we're pioneering this breakthrough technology to sustainably improve medicine for the benefit of all patients.



The scientific overlay on the title is not that of the individual pictured and is not from a device of Siemens Healthineers. It is modified for better visualization.

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